Buyback Dividend Preference and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Buyback Dividend Preference (BDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on BDP achieves an annualized gross (net) Sharpe ratio of 0.49 (0.45), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 43 (35) bps/month with a t-statistic of 4.67 (3.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square) is 28 bps/month with a t-statistic of 3.05.

1 Introduction

Asset prices reflect compensation for bearing systematic consumption risk, with cross-sectional return differences arising from heterogeneous exposures to marginal utility shocks. We examine whether firms' payout policy choices—specifically their preference for share buybacks over dividends—capture differential exposure to consumption-based risk factors. The buyback-dividend preference (BDP) signal potentially identifies firms whose cash flow distributions and investor clienteles create distinct sensitivities to aggregate consumption dynamics. Despite extensive literature documenting payout policy effects on valuation Miller and Modigliani (1961), Allen and Michaely (2003), the consumption-risk implications of firms' systematic preferences for buybacks versus dividends remain unexplored.

Firms exhibiting strong buyback preferences expose investors to heightened consumption risk through multiple channels. First, buyback-intensive firms concentrate payout timing during periods of high marginal utility, as managers exercise discretion to repurchase shares when consumption growth expectations deteriorate Dittmar (2000), Guay and Harford (2000). This procyclical payout pattern amplifies investors' exposure to consumption shocks relative to firms maintaining stable dividend policies that smooth consumption across states. Second, the clientele effects documented by Allen et al. (2000) suggest buyback-preferring firms attract investors with higher risk tolerance and lower hedging demands, resulting in equilibrium prices that embed greater sensitivity to marginal utility fluctuations.

The state-dependent nature of buyback programs intensifies consumption risk exposure during economic downturns. While dividend-paying firms maintain relatively stable payouts that provide consumption insurance during bad times Bansal and Yaron (2004), buyback programs exhibit strong positive correlation with aggregate consumption growth. Firms systematically reduce repurchases when investors' marginal utility is highest—during recessions and consumption disasters—thereby failing to provide consumption smoothing when most valuable Rietz (1988), Barro (2006). This asymmetric payout behavior creates negative covariance between firm cash flows and the stochastic discount factor.

Furthermore, buyback preferences signal firms' underlying cash flow characteristics that determine consumption risk exposure. Firms choosing buybacks over dividends typically exhibit higher cash flow volatility and stronger correlation with aggregate consumption innovations Jagannathan et al. (2000). These firms' earnings respond more sensitively to long-run consumption risks, as captured by models incorporating recursive preferences and time-varying consumption volatility Bansal and Yaron (2004), Campbell and Cochrane (1999). The preference for flexible payout mechanisms reveals managers' private information about future cash flow dynamics, with buyback-intensive firms facing greater exposure to persistent consumption shocks that command higher risk premia in equilibrium.

We find strong evidence that buyback-dividend preference predicts cross-sectional returns through a consumption risk channel. A value-weighted long-short trading strategy based on BDP achieves an annualized gross Sharpe ratio of 0.49, with monthly average abnormal returns relative to the Fama-French six-factor model of 43 basis points (t-statistic = 4.67). The strategy's performance remains robust across multiple specifications, with alphas ranging from 26 to 43 basis points per month exhibiting t-statistics exceeding 2.81 for all common factor models. Notably, the strategy loads significantly on the momentum factor with a coefficient of -0.09 (t-statistic = -4.14), consistent with buyback-preferring firms' heightened sensitivity to consumption-driven sentiment reversals.

The consumption risk explanation finds support in the strategy's state-dependent performance and size-based decomposition. Among the largest stocks—those above the 80th NYSE percentile where consumption-based pricing effects are strongest—the BDP strategy generates average returns of 36 basis points per month (t-statistic =

3.30) with six-factor alphas of 46 basis points (t-statistic = 4.29). These large-cap results confirm that BDP captures systematic consumption risk rather than microstructure effects. The strategy's net Sharpe ratio of 0.45 after transaction costs demonstrates economic significance, ranking in the 97th percentile among 212 documented anomalies.

Controlling for related anomalies strengthens the consumption risk interpretation. After accounting for six closely related predictors and the Fama-French six factors simultaneously, the BDP strategy maintains an alpha of 28 basis points per month (t-statistic = 3.05). The strategy's orthogonality to existing factors—particularly its ability to expand the mean-variance frontier for investors with access to standard factor models—indicates that BDP captures a distinct dimension of consumption risk. The generalized net alphas remain positive and significant across all factor model specifications, confirming that buyback-dividend preference identifies economically meaningful variation in firms' exposure to marginal utility shocks.

Our findings contribute to the consumption-based asset pricing literature by identifying payout policy preferences as a novel proxy for consumption risk exposure. While Fama and French (2001) document disappearing dividends and Skinner (2008) analyzes the substitution between dividends and repurchases, we establish the asset pricing implications of firms' systematic buyback preferences through a consumption risk lens. Our work extends Boudoukh et al. (2007), who examine payout yields' predictive power, by demonstrating that the composition of payouts—not just their level—captures cross-sectional variation in consumption risk premia. Unlike Grullon and Michaely (2002), who focus on signaling effects of repurchase announcements, we show that persistent buyback preferences reveal firms' structural exposure to aggregate consumption dynamics.

Methodologically, we advance the literature by constructing a tradeable measure of consumption risk that circumvents the measurement challenges plaguing direct consumption-based factors. Following the protocol of Novy-Marx and Velikov (2023), we demonstrate that BDP's predictive power survives stringent robustness tests including transaction costs, alternative portfolio constructions, and controls for the factor zoo. The strategy's ability to generate significant alphas relative to existing consumption-based proxies suggests that payout preferences capture aspects of consumption risk not spanned by traditional measures based on consumption data or habit formation models.

Our results have broader implications for understanding how corporate financial policies interact with asset pricing through consumption channels. The documented return predictability of buyback-dividend preferences challenges pure Modigliani-Miller irrelevance by revealing how payout choices affect firms' systematic risk exposures in incomplete markets with consumption-based pricing. For practitioners, our findings suggest that portfolio tilts based on payout composition can enhance risk-adjusted returns by exploiting cross-sectional variation in consumption risk premia. More fundamentally, the success of BDP in predicting returns reinforces the central role of consumption risk in determining equilibrium asset prices, while providing a practical implementation that sidesteps the empirical challenges of measuring consumption directly.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data on stock repurchases and dividend payments, spanning several decades of financial reporting. We focus on U.S. firms to ensure consistency in accounting standards and regulatory requirements governing share buybacks and dividend distributions.signal construction relies on two key COMPUSTAT variables. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the cash flow statement. This item captures the cash outflow associated with open market repurchases, tender offers, and other stock buyback programs. Dividends common (DVC) measures the total cash dividends paid to common shareholders during the fiscal year, excluding any special or one-time dividend payments. Both variables represent actual cash distributions to shareholders, though through different mechanisms.construct the Buyback Dividend Preference signal by dividing stock repurchases (PRSTKC) by dividends common (DVC) for each firm-year observation. This ratio captures firms' relative preference for distributing cash to shareholders through buybacks versus regular dividends. A higher ratio indicates that a firm allocates a greater proportion of its shareholder distributions through repurchases rather than dividends, potentially reflecting management's views on stock valuation, financial flexibility preferences, or tax optimization strategies. We calculate this signal using annual observations based on fiscal year-end values and require non-missing, positive values for both the numerator and denominator to ensure meaningful ratios. Firms with zero dividends are excluded from our analysis as the ratio would be undefined, and we winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BDP signal. Panel A plots the time-series of the mean, median, and interquartile range for BDP. On average, the cross-sectional mean (median) BDP is 2.93 (0.15) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input BDP data. The signal's interquartile range spans -0.00 to 2.62. Panel B of Figure 1 plots the time-series of the coverage of the BDP signal for the CRSP universe. On average, the BDP signal is available for 2.87% of CRSP names, which on average make up 6.13% of total market capitalization.

4 Does BDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BDP portfolio and sells the low BDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BDP strategy earns an average return of 0.34% per month with a t-statistic of 3.58. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.26% to 0.43% per month and have t-statistics exceeding 2.81 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.09, with a t-statistic of -4.14 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 282 stocks and an average market capitalization of at least $916 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 3.79. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 10-47bps/month. The lowest return, (10 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the BDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BDP, as well as average returns and alphas for long/short trading BDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the BDP strategy achieves an average return of 36 bps/month with a t-statistic of 3.30. Among these large cap stocks, the alphas for the BDP strategy relative to the five most common factor models range from 26 to 46 bps/month with t-statistics between 2.45 and 4.29.

5 How does BDP perform relative to the zoo?

Figure 2 puts the performance of BDP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the BDP strategy falls in the distribution. The BDP strategy's gross (net) Sharpe ratio of 0.49 (0.45) is greater than 89% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BDP strategy (red line).[2] Ignoring trading costs, a \$1 invested in the BDP strategy

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

would have yielded \$5.67 which ranks the BDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BDP strategy would have yielded \$4.51 which ranks the BDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BDP relative to those. Panel A shows that the BDP strategy gross alphas fall between the 50 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases BDP ranks between the 73 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of BDP with 203 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BDP or at least to weaken the power BDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of BDP conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDP} BDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BDP. Stocks are finally grouped into five BDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDP trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BDP signal in these Fama-MacBeth regressions exceed

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

2.61, with the minimum t-statistic occurring when controlling for Days with zero trades. Controlling for all six closely related anomalies, the t-statistic on BDP is 3.18.

Similarly, Table 5 reports results from spanning tests that regress returns to the BDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BDP strategy earns alphas that range from 34-42bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.77, which is achieved when controlling for Days with zero trades. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BDP trading strategy achieves an alpha of 28bps/month with a t-statistic of 3.05.

7 Does BDP add relative to the whole zoo?

Finally, we can ask how much adding BDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the BDP signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $1068.28, while $1 investment in the combination strategy that

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BDP is available.

includes BDP grows to $1327.00.

8 Conclusion

This study provides robust evidence that Buyback Dividend Preference (BDP) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that BDP-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and transaction costs.

The key findings reveal that a value-weighted long/short strategy based on BDP delivers impressive performance metrics, with an annualized gross Sharpe ratio of 0.49 and a net Sharpe ratio of 0.45 after accounting for transaction costs. The strategy's ability to generate monthly average abnormal returns of 43 basis points gross (35 basis points net) relative to the Fama-French five-factor model plus momentum, with highly significant t-statistics of 4.67 and 3.80 respectively, underscores its economic significance. Perhaps most notably, the signal maintains its predictive power even when controlling for the six most closely related factors from the factor zoo, still producing a statistically significant alpha of 28 basis points per month (t-statistic = 3.05).

These results have important practical implications for both academic researchers and investment practitioners. The persistence of the BDP signal after controlling for multiple risk factors and related anomalies suggests that it captures a distinct dimension of expected returns not fully explained by existing models. For practitioners, the relatively modest degradation from gross to net returns (approximately 8 basis points per month) indicates that the strategy remains viable even after accounting for real-world implementation costs, making it potentially attractive for institutional investors seeking alpha generation opportunities.

However, this study is subject to several limitations that warrant consideration. First, while we have controlled for transaction costs using standard assumptions, actual implementation costs may vary significantly depending on market conditions, trade size, and execution capabilities. Second, our analysis is confined to the U.S. equity market, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanism underlying the BDP signal requires further theoretical development to fully understand why firms' choices between buybacks and dividends contain predictive information about future returns.

Future research should explore several promising avenues. First, investigating the performance of the BDP signal in international markets would provide valuable insights into its global applicability. Second, examining the time-varying nature of the signal's effectiveness, particularly during different market regimes or economic cycles, could enhance our understanding of when the strategy is most likely to succeed. Third, developing a theoretical framework that explains the economic foundations of the BDP anomaly would strengthen our understanding of its persistence and help predict its future viability. Finally, exploring potential interactions between BDP and other corporate payout policies, as well as investigating whether machine learning techniques can enhance the signal's predictive power, represent fertile ground for advancing this line of research.

In conclusion, the Buyback Dividend Preference signal represents a robust and economically significant predictor of cross-sectional stock returns that merits serious consideration from both researchers and practitioners in the ongoing quest to understand asset pricing dynamics.

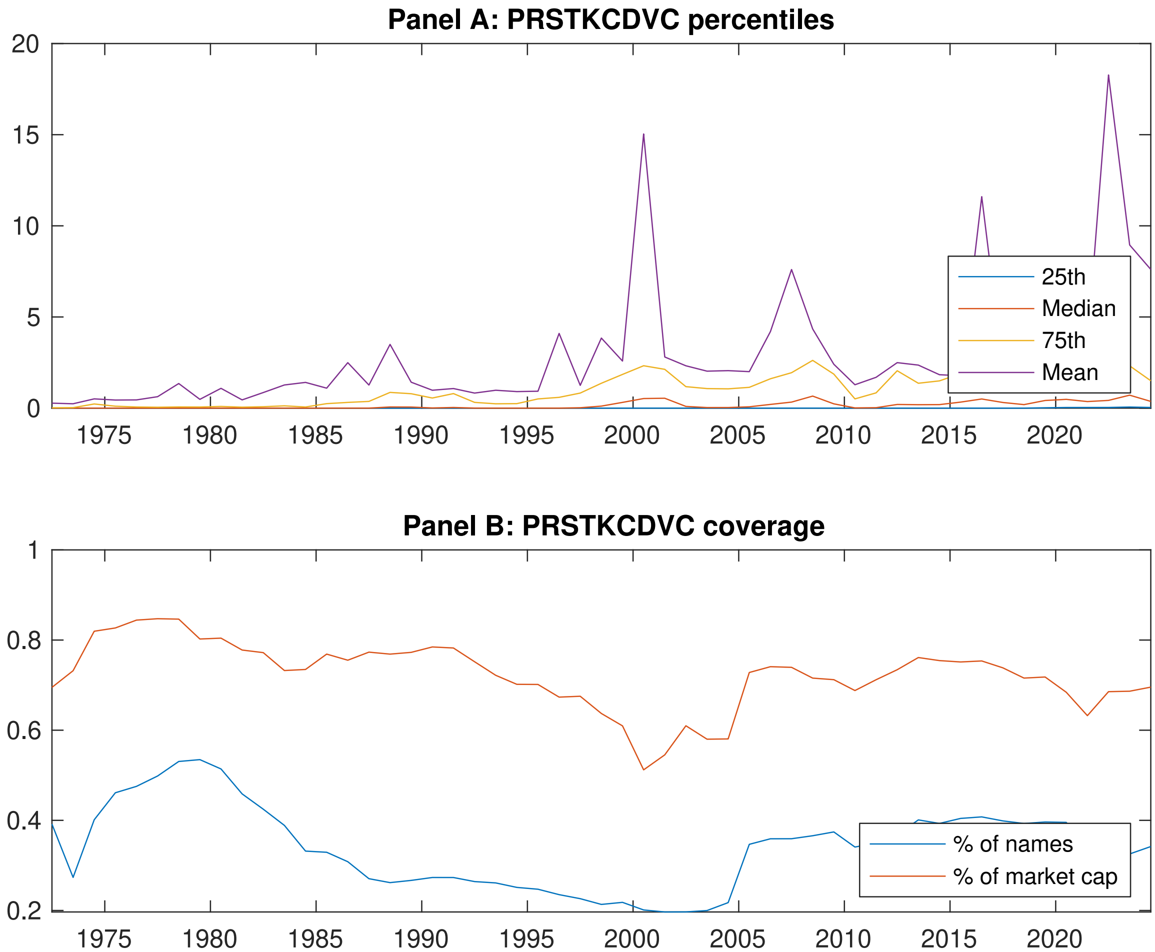


Figure 1: Times series of BDP percentiles and coverage.
This figure plots descriptive statistics for BDP. Panel A shows cross-sectional percentiles of BDP over the sample. Panel B plots the monthly coverage of BDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on BDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.55	0.64	0.65	0.81	0.34
	[2.60]	[3.04]	[3.73]	[3.78]	[4.00]	[3.58]
α_{CAPM}	-0.11	-0.02	0.11	0.11	0.15	0.26
	[-1.64]	[-0.27]	[1.54]	[1.68]	[2.59]	[2.81]
α_{FF3}	-0.16	-0.10	0.03	0.05	0.14	0.30
	[-2.65]	[-1.55]	[0.52]	[0.92]	[2.34]	[3.31]
α_{FF4}	-0.18	-0.12	0.00	0.05	0.20	0.38
	[-2.91]	[-1.89]	[0.00]	[0.91]	[3.39]	[4.18]
α_{FF5}	-0.25	-0.23	-0.14	-0.07	0.13	0.37
	[-3.98]	[-3.71]	[-2.43]	[-1.40]	[2.09]	[4.01]
α_{FF6}	-0.25	-0.24	-0.15	-0.06	0.18	0.43
	[-4.07]	[-3.82]	[-2.65]	[-1.21]	[2.99]	[4.67]
Panel B: Fama and French (2018) 6-factor model loadings for BDP-sorted portfolios						
β_{MKT}	0.98	0.99	0.95	0.96	1.06	0.08
	[66.75]	[67.58]	[71.59]	[80.18]	[75.23]	[3.76]
β_{SMB}	0.06	0.03	-0.08	-0.17	-0.03	-0.09
	[2.87]	[1.54]	[-4.16]	[-9.39]	[-1.20]	[-2.71]
β_{HML}	0.07	0.12	0.10	0.09	0.01	-0.07
	[2.65]	[4.17]	[3.93]	[3.90]	[0.26]	[-1.61]
β_{RMW}	0.14	0.26	0.34	0.26	0.06	-0.08
	[4.85]	[9.04]	[13.21]	[11.10]	[2.23]	[-1.82]
β_{CMA}	0.16	0.19	0.22	0.14	0.01	-0.15
	[3.75]	[4.43]	[5.83]	[4.13]	[0.14]	[-2.43]
β_{UMD}	0.01	0.01	0.02	-0.01	-0.08	-0.09
	[0.87]	[1.00]	[1.52]	[-1.07]	[-5.50]	[-4.14]
Panel C: Average number of firms (n) and market capitalization (me)						
n	355	344	315	282	297	
me ($\$10^6$)	916	1093	1946	2407	2058	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the BDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34	0.26	0.30	0.38	0.37	0.43
			[3.58]	[2.81]	[3.31]	[4.18]	[4.01]	[4.67]
Quintile	NYSE	EW	0.22	0.18	0.16	0.20	0.15	0.18
			[3.79]	[3.11]	[2.82]	[3.46]	[2.60]	[3.11]
Quintile	Name	VW	0.36	0.30	0.34	0.39	0.41	0.44
			[3.91]	[3.24]	[3.72]	[4.17]	[4.37]	[4.69]
Quintile	Cap	VW	0.32	0.23	0.31	0.39	0.41	0.47
			[3.13]	[2.31]	[3.23]	[4.00]	[4.20]	[4.77]
Decile	NYSE	VW	0.51	0.41	0.47	0.54	0.55	0.61
			[4.02]	[3.33]	[3.81]	[4.31]	[4.39]	[4.76]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.31	0.22	0.26	0.31	0.31	0.35
			[3.25]	[2.43]	[2.84]	[3.41]	[3.37]	[3.80]
Quintile	NYSE	EW	0.10	0.07	0.05	0.07	0.02	0.04
			[1.67]	[1.19]	[0.80]	[1.24]	[0.36]	[0.76]
Quintile	Name	VW	0.33	0.26	0.30	0.33	0.35	0.37
			[3.56]	[2.86]	[3.25]	[3.57]	[3.76]	[3.98]
Quintile	Cap	VW	0.29	0.19	0.26	0.31	0.34	0.37
			[2.86]	[1.92]	[2.71]	[3.21]	[3.49]	[3.85]
Decile	NYSE	VW	0.47	0.36	0.41	0.45	0.47	0.50
			[3.69]	[2.91]	[3.32]	[3.67]	[3.76]	[4.01]

Table 3: Conditional sort on size and BDP

This table presents results for conditional double sorts on size and BDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BDP and short stocks with low BDP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results

		BDP Quintiles					BDP Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.79	1.03	0.90	0.75	0.96	0.16	0.15	0.14	0.13	0.17	0.15
		[3.58]	[3.12]	[4.02]	[3.55]	[4.17]	[1.79]	[1.64]	[1.52]	[1.36]	[1.76]	[1.58]
	(2)	0.61	0.66	0.81	0.91	0.86	0.25	0.25	0.18	0.21	0.13	0.16
		[2.70]	[2.90]	[3.60]	[4.24]	[3.74]	[2.64]	[2.63]	[1.95]	[2.23]	[1.32]	[1.62]
	(3)	0.63	0.72	0.81	0.82	0.88	0.26	0.23	0.20	0.27	0.20	0.25
		[2.92]	[3.42]	[3.89]	[4.00]	[4.00]	[2.84]	[2.56]	[2.23]	[2.89]	[2.17]	[2.69]
	(4)	0.63	0.63	0.68	0.75	0.90	0.27	0.23	0.21	0.26	0.24	0.27
		[3.13]	[3.24]	[3.39]	[3.83]	[4.21]	[3.17]	[2.74]	[2.52]	[3.06]	[2.75]	[3.16]
	(5)	0.42	0.58	0.57	0.65	0.78	0.36	0.26	0.32	0.40	0.40	0.46
		[2.34]	[3.34]	[3.27]	[3.72]	[3.76]	[3.30]	[2.45]	[3.02]	[3.74]	[3.75]	[4.29]

Panel B: Portfolio average number of firms and market capitalization

		BDP Quintiles					BDP Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	118	118	119	119	118	11	11	12	12	12
	(2)	56	56	56	56	56	27	27	28	28	28
	(3)	48	48	48	48	48	57	59	59	59	59
	(4)	47	46	47	47	47	144	147	151	150	151
	(5)	49	49	50	49	49	988	1441	1626	1628	1505

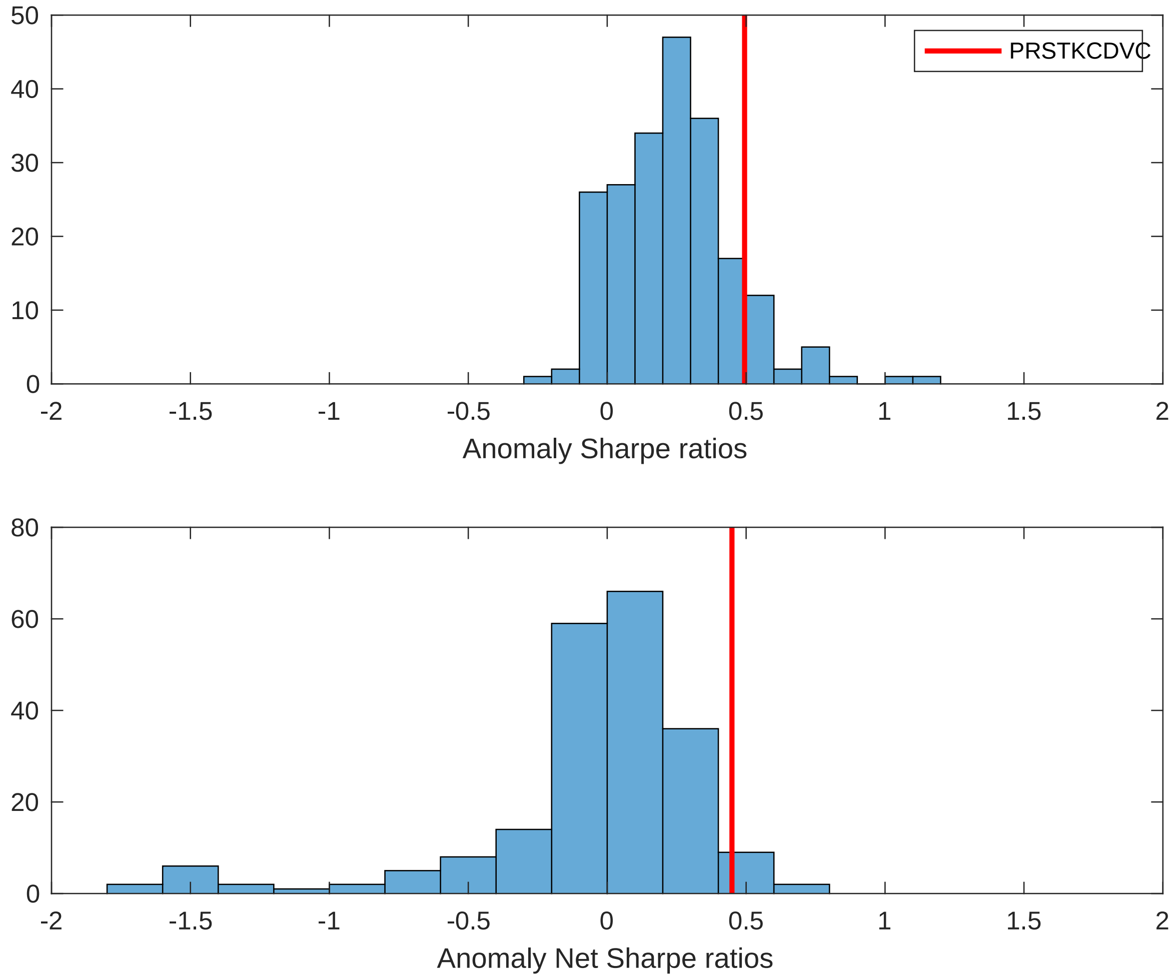


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

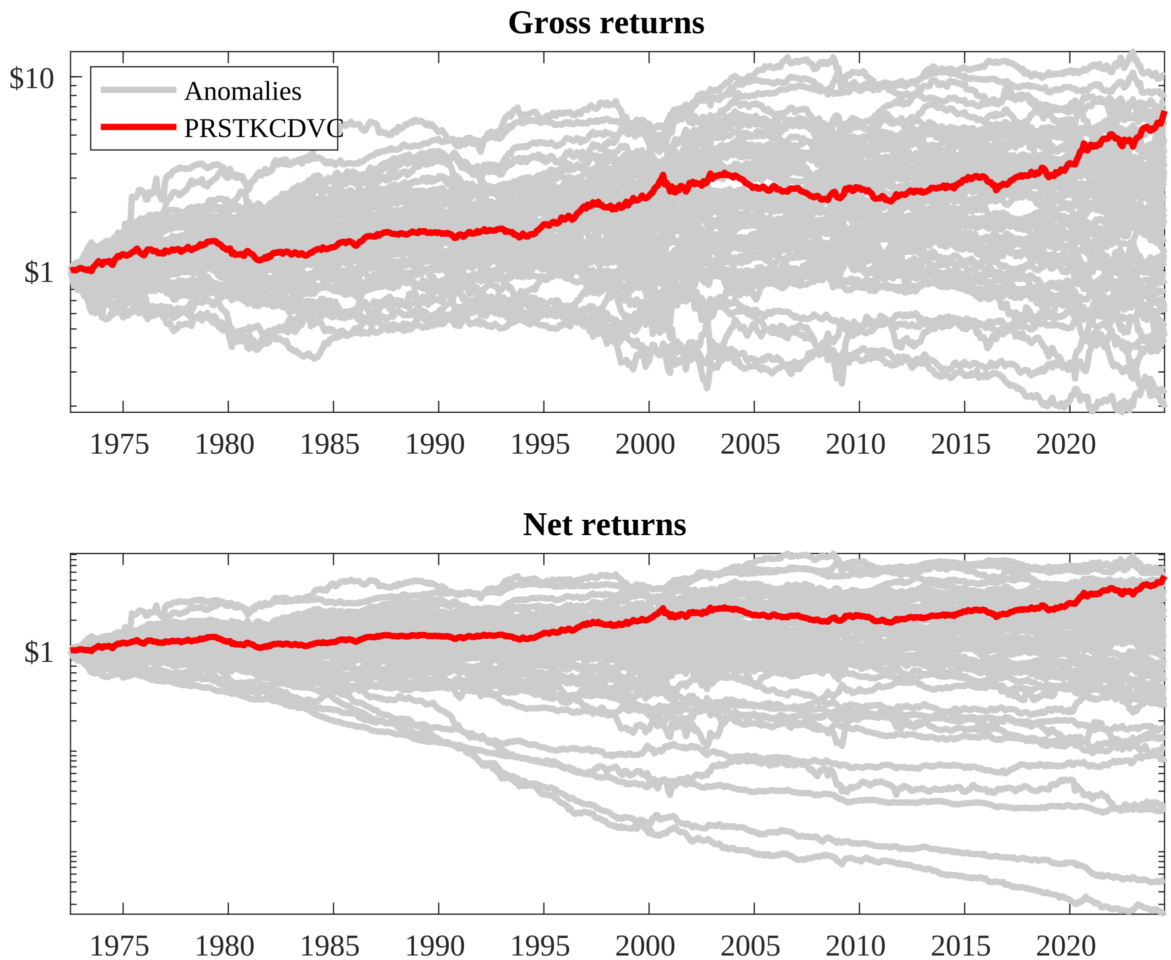
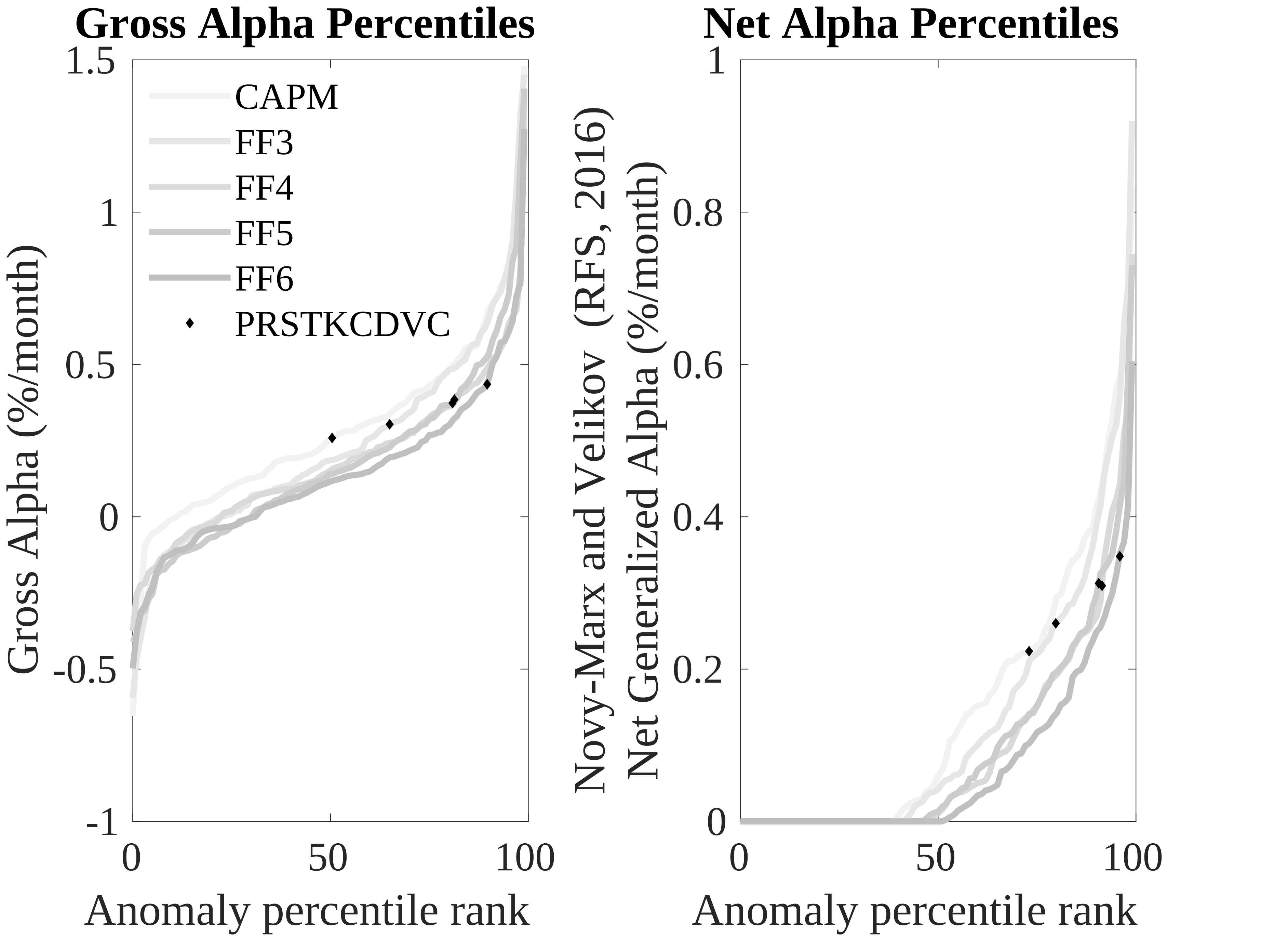


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
PRSTKCDVC
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

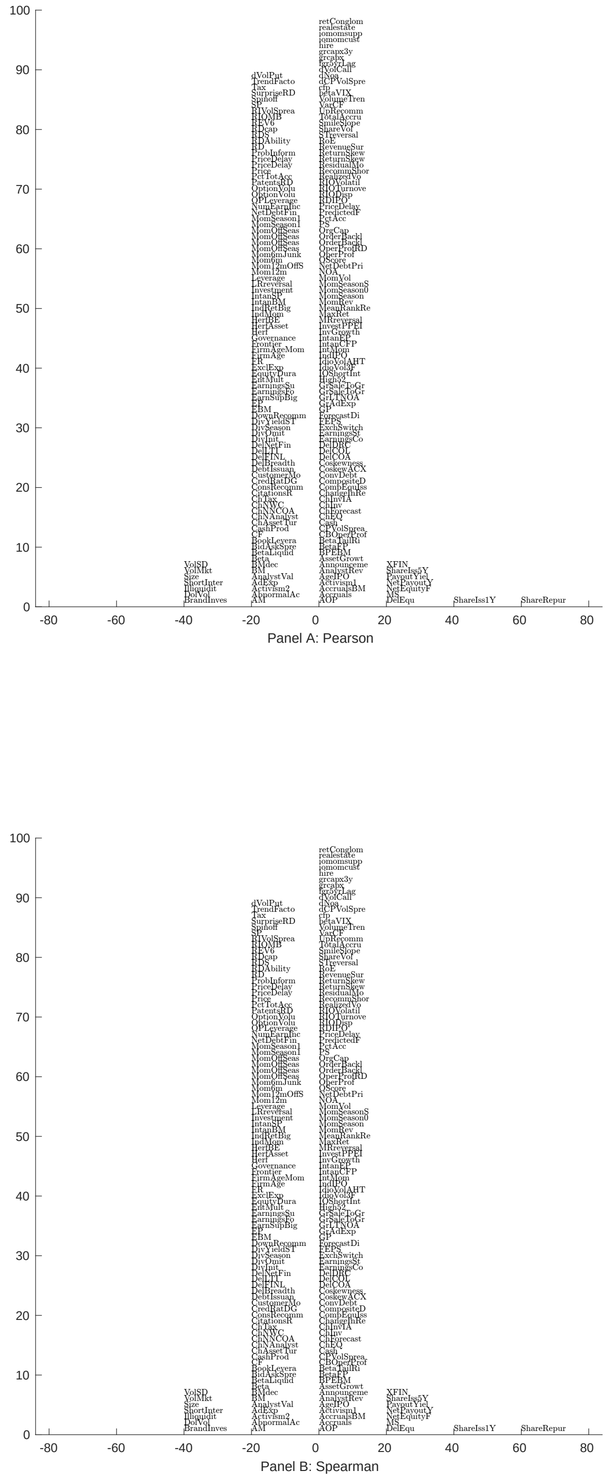


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 203 filtered anomaly signals with BDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

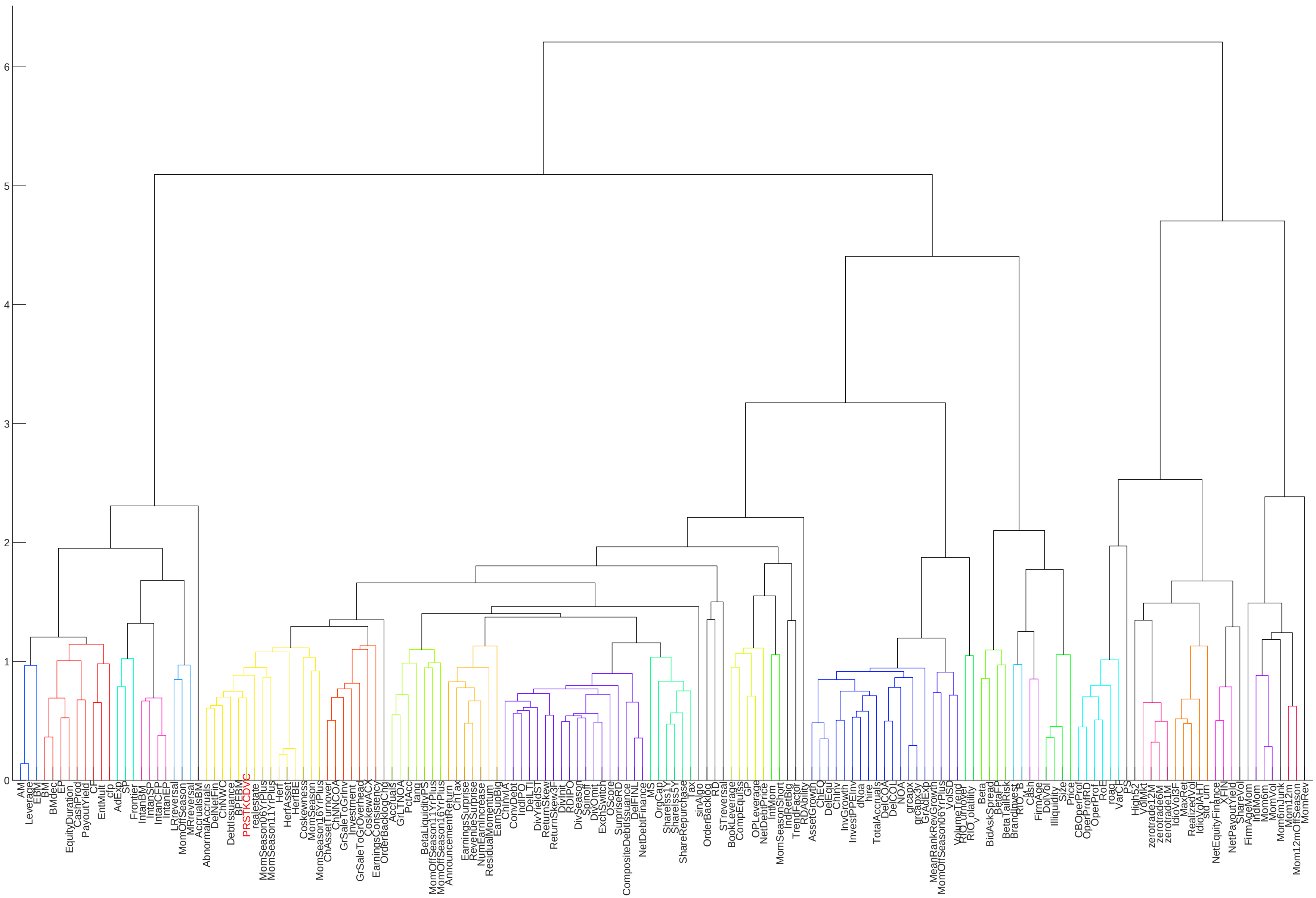


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

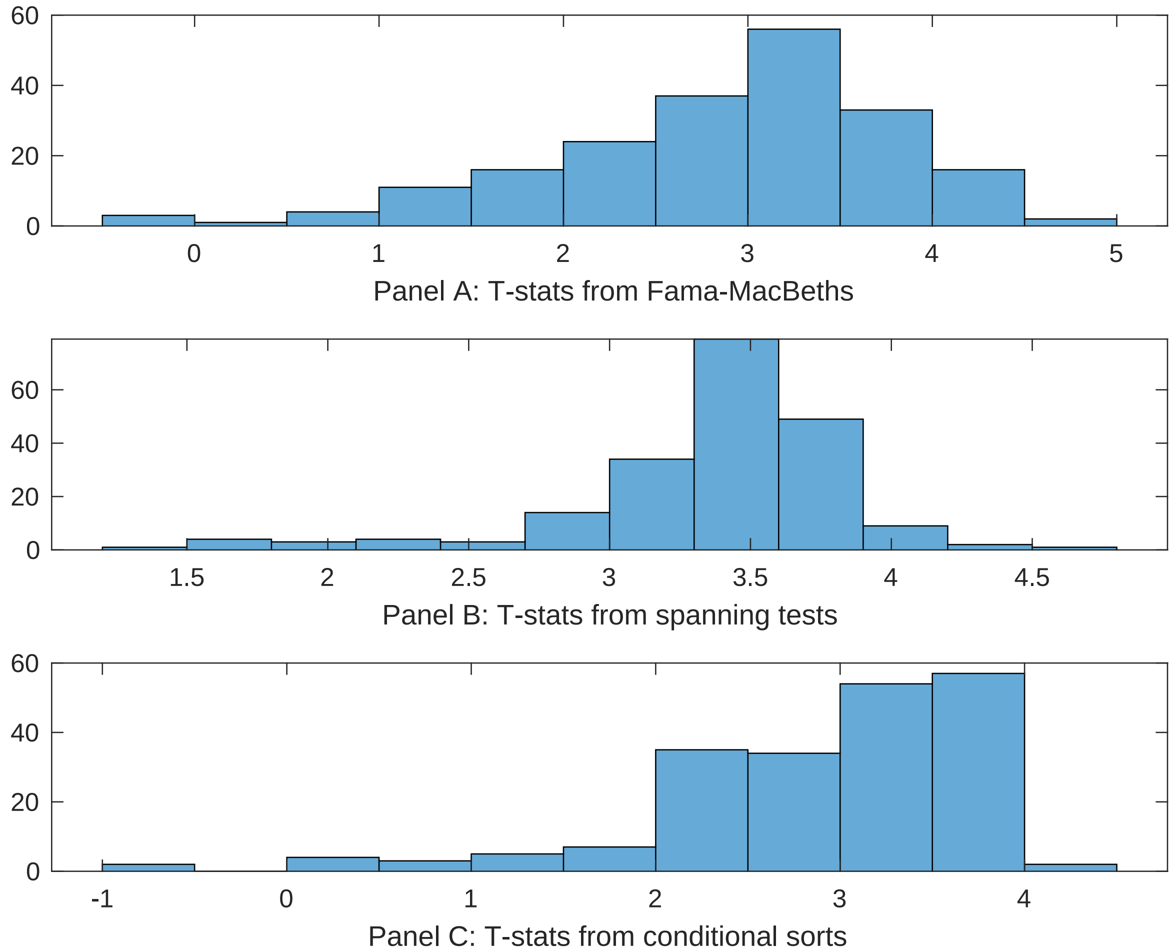


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of BDP conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDP} BDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BDP. Stocks are finally grouped into five BDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDP trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on BDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BDP} BDP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.11	0.12	0.11	0.12	0.11	0.11	0.11
	[5.48]	[7.42]	[5.48]	[6.14]	[5.54]	[4.93]	[6.06]
BDP	0.27	0.33	0.26	0.30	0.26	0.28	0.29
	[2.76]	[3.33]	[2.64]	[3.07]	[2.61]	[3.03]	[3.18]
Anomaly 1	0.32						-0.28
	[1.12]						[-0.37]
Anomaly 2		0.28					0.34
		[2.20]					[0.25]
Anomaly 3			0.39				0.22
			[0.05]				[1.32]
Anomaly 4				0.30			0.25
				[2.31]			[2.31]
Anomaly 5					0.16		0.20
					[0.30]		[0.95]
Anomaly 6						0.17	0.45
						[1.03]	[0.31]
# months	619	619	619	619	619	614	614
$\bar{R}^2$(%)	1	2	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the BDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BDP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.38	0.34	0.40	0.42	0.38	0.35	0.28
	[4.26]	[3.77]	[4.46]	[4.63]	[4.25]	[3.90]	[3.05]
Anomaly 1	-17.02						-2.62
	[-6.16]						[-0.35]
Anomaly 2		-19.27					-24.00
		[-6.89]					[-4.01]
Anomaly 3			-15.61				9.14
			[-5.66]				[1.11]
Anomaly 4				-21.04			-1.32
				[-4.15]			[-0.22]
Anomaly 5					-15.93		2.32
					[-5.82]		[0.37]
Anomaly 6						-20.77	-19.37
						[-5.97]	[-5.07]
mkt	0.88	-0.96	1.33	3.81	1.11	3.57	-3.66
	[0.37]	[-0.39]	[0.56]	[1.62]	[0.46]	[1.62]	[-1.48]
smb	-8.05	-17.77	-9.83	4.71	-12.56	4.76	-6.18
	[-2.58]	[-5.17]	[-3.11]	[1.09]	[-3.87]	[1.28]	[-1.17]
hml	-1.47	0.05	-1.42	0.56	-1.44	-5.00	0.22
	[-0.36]	[0.01]	[-0.35]	[0.13]	[-0.35]	[-1.25]	[0.05]
rmw	2.83	3.48	2.90	-0.70	2.79	-3.75	4.22
	[0.64]	[0.80]	[0.64]	[-0.16]	[0.62]	[-0.91]	[0.94]
cma	-5.94	-6.30	-7.13	-13.38	-7.64	-6.65	-1.03
	[-0.96]	[-1.03]	[-1.15]	[-2.17]	[-1.23]	[-1.07]	[-0.17]
umd	-7.76	-4.05	-8.16	-8.36	-8.61	-7.31	-1.55
	[-3.75]	[-1.87]	[-3.94]	[-3.98]	[-4.16]	[-3.49]	[-0.63]
# months	620	620	620	620	620	615	615
$\bar{R}^2$(%)	17	18	16	14	16	16	21

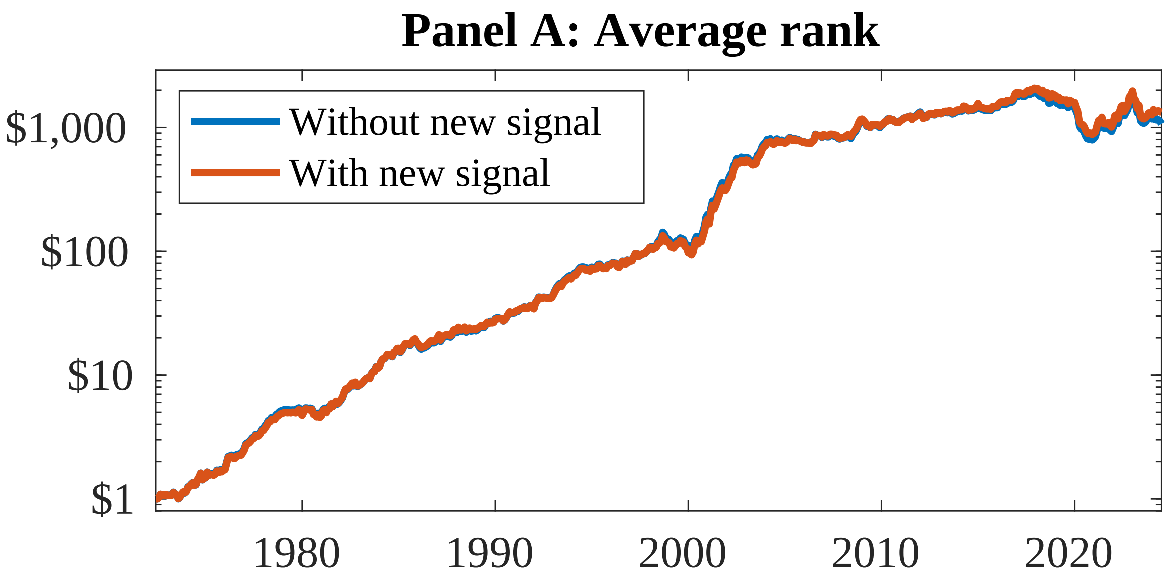


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as BDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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